STAFF REVIEW · H446 024 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Do Now — source answers

1. The most recently added item (LIFO).

2. The earliest added item (FIFO).

3. Pointers/indexes.

4. Create, traverse, add data, remove data.

I Do — source worked steps

1. Name stack rule — CODE:
Stack removes the most recently added item first.

2. Name queue rule — CODE:
Queue removes the earliest added item first.

3. Link the two — CODE:
Therefore the same input order produces different removal orders.

4. Use exam terms — CODE:
LIFO versus FIFO

We Do — guided CHECK

1. Improve the answer
CODE:
An array has a fixed size once created, whereas a list can grow or shrink.

2. Improve the answer
CODE:
A tuple differs because ...
CODE:
A tuple is fixed / cannot be changed in the same way, whereas a list is mutable and can change.

3. Improve the answer
CODE:
Use the ? pointer to locate the first item, then ...
CODE:
Use the start pointer to locate the first item, read its next pointer, then use that pointer value as the index of the second item.

4. Improve the answer
CODE:
Left/right contain ...
CODE:
Left/right contain pointer/index values to child rows; leaf pointers are null.

You Do — source mark schemes

1. Compare structures
CODE:
Stack vs queue:
A stack is LIFO; a queue is FIFO.

Array vs list:
An array has a fixed size once created; a list can grow/shrink.

Tuple vs list:
A tuple is fixed/immutable in structure whereas a list can be changed.

Linked-list second item:
Use the start pointer to locate the first item. Read that item's next pointer and use its value as the index/address of the second item.
Check: Use precise structure vocabulary and relational wording.

2. BST array representation
CODE:
0: 1 | Sligo | 3
1: 2 | Dublin | 4
2: null | Cork | null
3: 6 | Waterford | null
4: 7 | Galway | 5
5: null | Limerick | null
6: null | Tralee | null
7: null | Dundalk | null
Check: Use pointer indexes in Left/Right and null for leaf pointers.

3. Record + BST
CODE:
Record:
It groups several related fields about one task into a single structured item, while allowing the fields to have different data types.

BST characteristics:
Each node has at most two children.
Values/keys smaller than a node are placed to the left and larger values/keys to the right.
Check: Link each answer directly to the context.

4. Create / traverse / add / remove
CODE:
Linked list add: update pointers so the new item's next points to the following item.
Stack add/remove: push/pop at the top.
Queue add/remove: enqueue at rear, dequeue at front.
Graph add/remove: set/clear adjacency-matrix entry or entries.
BST add: follow comparisons until a null child pointer and attach the new index.
Hash table add: hash the key, then place it in the bucket using the collision strategy.
Check: Your responses describe the structural change, not just name the structure.

Exit Ticket — source answer

1. True. In a pointer-based representation of a BST, left and right fields identify child nodes, not copies of the node labels.